

Introduction to the analysis of UCEs Version 1: Getting Started: HPC Cluster Access, File Management, Environment Setup, and Software Installation

Abstract

This tutorial is the first part of the **Introduction to the Analysis of UCEs V1** guide. In this practice notebook, you will learn how to install the programs required terminal for bioinformatics work in phylogenomics using ultraconserved elements (UCEs).

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The texts presented here were written by FG, with English language corrections made with the assistance of ChatGPT. The commands are adaptations developed by FGarcía with the help of IAshton and JGonzales, based on tutorials and the official documentation of the tools used. It is strongly recommended to consult the official guides of each program for a deeper understanding.

Practice 1: Installing WSL (Windows Subsystem for Linux) on Windows 10/11

WSL (Windows Subsystem for Linux) is a tool that allows you to use Linux inside Windows, without needing to install a virtual machine, use dual boot, or modify disk partitions. It works as a special system layer that translates Linux instructions so Windows can execute them quickly and efficiently. It is a practical way to learn and use Linux without leaving Windows, fully compatible with most terminal-based programs and perfect for students beginning in bioinformatics or programming.

Important: To install or activate WSL, you must open PowerShell as Administrator. Without administrator permissions, the commands will not work.

1. Check if WSL is already installed

```
wsl --status
```

- If WSL is installed, you will see version and distribution information.
- If it is not installed, Windows will display a message indicating that no distribution is available.

2. Install WSL

```
wsl --install
```

This command automatically:

- Enables the required components (WSL and Virtual Machine Platform)
- Installs the latest version of WSL
- Installs **Ubuntu** as the default distribution

After the process finishes, Windows will ask you to **restart** your computer.

You can check the status again with:

3. Activate and launch WSL

To start WSL and enter Linux, simply run:

```
wsl
```

Now you have WSL installed on your machine and you can use Linux directly inside Windows.

On MacBook, this step is not necessary, WSL is only for Windows 10/11 users.

Practice 2. Accessing the SDSU server

In bioinformatics, we work with large datasets, and many analyses—such as genome assemblies, large-scale alignments, or phylogenomic inference—can take hours or even days to complete. Personal computers, which usually have 4 to 16 CPU cores, are not sufficient for these tasks and can even be damaged under such heavy computational load.

For this reason, we use computing clusters, which are groups of interconnected nodes equipped with multiple CPUs, large amounts of RAM, and resource managers that allow analyses to run efficiently in parallel.

At SDSU, you can use the Mesxuuyan cluster, for which you will receive a username and password. In this tutorial, you will learn how to log in and start working; most clusters operate in a very similar way.

1. Log in to the cluster using your username. After that, it will prompt you for your password — just type your password when asked.

```
ssh yourusername@mesx.sdsu.edu  
Password: ?
```

2. When you log in, you first land on a “front-end node.” This is like the entry point of the cluster. It's not where the heavy computation happens — it's mainly for logging in, organizing files, and submitting jobs.

```
***** NOTICE *****  
*****  
Welcome to a MESX front-end node.  
  
TO connect to MESX proper do,  
  
$ rsh mesx  
or  
$ ssh mesx  
*****  
  
***** end NOTICE *****  
  
Activate the web console with: systemctl enable --now cockpit.socket  
Last login: Tue Dec 2 15:53:05 2025 from 146.244.34.130  
(base) [fgarcia@mesxuuyan ~]$ rsh mesx
```

```
rsh mesx
```

3. After running `rsh mesx`, you will see a notice explaining that MESX is a batch-only HPC cluster. This means you should not run programs directly on the head node (`mesx`). The head node is only for preparing your work, and all computations must be submitted through the PBS/Torque job scheduler. Your jobs will then be sent to one of the 20 compute nodes. You can check the `/examples` folder for sample job scripts.
4. On this cluster, each user has their own scratch directory, which is the place where you should store your data and run your analyses. Scratch directories are designed for large files and high-performance computing, unlike the `/home` folder, which is small and meant only for basic settings or scripts. When your account was created, the system assigned you a specific scratch path (for example: `/scratch2/userdata2/username`).

You received this path in the email with your access information. This folder is your workspace on the cluster — it is where you must put your datasets, results, scripts, and everything related to your analysis.

```
cd /usr/scratch2/userdata2/fgarcia
```

Excellent! Now you know how to log in to the cluster. The next step is to get familiar with the environment and learn how to install the programs you need.

Finished a project? Make sure to clean up the files you generated or saved on the cluster. This helps free up space for yourself and for other users.

Always keep your credentials private — never share your username or password with anyone. Also, avoid posting screenshots, photos, or the cluster's address online. Following these simple precautions keeps the system safe, organized, and efficient for everyone!

Note: If you are logging in from outside the SDSU Wi-Fi network, you must use the SDSU VPN. For more information and setup instructions, please check:

Practice 3. Installing Bioinformatics Tools with Conda

In bioinformatics, many analyses require specialized software to process, clean, and assemble sequencing data. Installing these programs manually can be complicated due to dependencies and version conflicts. **Conda** is a tool that solves this problem by managing packages and environments.

What is Conda?

Conda is a package and environment manager. It allows you to:

- Install software and all its dependencies automatically.
- Create isolated environments, so different projects can use different versions of the same program without conflicts.
- Work entirely from the terminal, making it ideal for clusters and remote servers.

Note: If you are using the SDSU cluster, **Conda is already installed**, and you do not need to install it. If you are using your own computer, you will need to install Conda first.

1. Create the Bioinfo environment

This creates a new isolated environment called Bioinfo.

The `-y` flag automatically confirms the creation.

Why a separate environment? Using an environment isolates the tools and dependencies from the rest of your system, avoiding conflicts between packages.

```
conda create -n Bioinfo -y
```

2. Activate the environment

```
conda activate Bioinfo
```

After activation, any packages you install will only exist inside the Bioinfo environment.

3. Install Trimmomatic and SPAdes

```
conda install -c bioconda trimmomatic spades fastqc -y
```

```
(bioinfo) ideobisium@Xorilbia ~ % conda install -c bioconda trimmomatic
2 channel Terms of Service accepted
Retrieving notices: done
Channels:
- bioconda
- conda-forge
- defaults
- faircloth-lab
Platform: osx-arm64
Collecting package metadata (repodata.json): /
```

The `-c bioconda` option tells conda to install packages from **Bioconda**, a specialized channel for bioinformatics software. Bioconda maintains a curated repository of popular bioinformatics tools, making installation and dependency management easier.

4. Verify the installations

```
spades.py --version
```

```
trimmomatic -version
```

```
(bioinfo) ideobisium@Xorilbia ~ % trimmomatic -version
spades.py --version
0.40
SPAdes genome assembler v4.2.0
(bioinfo) ideobisium@Xorilbia ~ %
```

- With `conda install -c bioconda <program>`, you can install most bioinformatics software available in Bioconda.
- Some programs may have newer versions that are **not yet on Bioconda**. In these cases, you might need to:
 - Download the binaries manually, or
 - Clone the repository from **GitHub** and build the program locally.

Since the SPAdes version available in Bioconda is outdated, we will install a newer version manually inside your bioinfo environment. The steps below explain what is happening and why each step is necessary.

1. Go to your personal folder

```
cd ~
```

This ensures that all the installation files and environments are stored in your personal space, avoiding system folders.

2. Download a compatible installer

```
wget  
https://github.com/ablab/spades/releases/download/v4.2.0/SPAdes-4.2.0-Linux.tar.gz
```

This command downloads a recent, precompiled SPAdes release that works on most Linux-based clusters.

3. Unzip (extract) the package

```
tar -xzf SPAdes-4.2.0-Linux.tar.gz
```

This extracts the SPAdes folder into your home directory so you can access the program files.

4. Add SPAdes to your PATH

```
echo 'export PATH=~/.SPAdes-4.2.0-Linux/bin:$PATH' >> ~/.bashrc  
source ~/.bashrc
```

This step makes the SPAdes executable available from anywhere in the terminal. By adding SPAdes to your PATH, you don't need to type the full directory every time — the system now knows where to find the program.

5. Check the installation

```
spades.py --version
```

If everything is correct, this will show the installed SPAdes version, confirming that the manual installation worked.

Some software packages may require additional configuration or dependency installation. These situations are explained in more detail in Practical 4 or 5.

Practice 4. Installing Miniconda3

Previously, we learned how to install bioinformatics packages using **Bioconda** with Conda. Most programs can be installed this way easily.

However, some tools require **Miniconda3**, a lightweight version of Conda.

What is Miniconda3?

Miniconda3 is like a **blank toolbox**: it comes with only the essential tools (Python and Conda), and you add only the tools you need. In contrast, Anaconda is like a **fully stocked toolbox** with hundreds of tools, many of which you may not use.

Using Miniconda3 allows us to **create clean, isolated environments** tailored for specific programs, like Phyluce, without conflicts or unnecessary software.

Check the documentation of MiniConda3 [here](#).

CLUSTER SDSU

If you are working on the SDSU Cluster, there is a problem with the Python version. You cannot install or change anything in the global cluster (rsx), only in your personal folder. For this reason, we need a previous version of Miniconda3 that is compatible with the Python version provided by Conda in the terminal.

1. Go to your personal folder

```
cd ~
```

This ensures that all the installation files and environments are stored in your personal space, avoiding system folders.

2. Download a compatible installer

```
wget https://repo.anaconda.com/miniconda/Miniconda3-py38_4.10.3-Linux-x86_64.sh
```

This downloads a stable version of Miniconda3 with Python 3.8, compatible with CentOS 7 (common on clusters).

3. Run the installer

```
bash Miniconda3-py38_4.10.3-Linux-x86_64.sh
```

The installer will ask you to read and accept the license. When prompted for the installation directory, enter a path inside your home folder, for example:

```
/home3/ashton/miniconda3
```

4. Initialize your shell

After installation, run:

```
source ~/.bashrc
```

This reloads your shell configuration so that your terminal knows about your new Miniconda installation.

5. Verify your Miniconda installation

```
which conda
```

This should show something like:

```
/home3/ashton/miniconda3/bin/conda
```

This confirms that you are using **your personal Miniconda**, not the system version.

Practice 5. Installing Phyluce

Phyluce is the main program for working in phylogenomics using UCEs. It requires Miniconda3 to be installed.

If you simply run: `conda install phyluce` (NO RUN IT)

you will get one of the very first versions of Phyluce, which lacks many of the current functions. Therefore, it is recommended to install a specific version directly from GitHub:

<https://github.com/faircloth-lab/phyluce/releases/tag/v1.7.3>

Inside your home folder, run the following commands.

1. Download the environment file for Linux:

```
wget  
https://raw.githubusercontent.com/faircloth-lab/phyluce/v1.7.3/distrib/phyluce-1.7.3-py36-Linux-conda.yml
```

2. Create a new Conda environment and install Phyluce:

```
conda env create -n phyluce-1.7.3 --file phyluce-1.7.3-py36-Linux-conda.yml
```

This will create a new environment named `phyluce-1.7.3` with all required dependencies. We are using the **Linux version**, since your terminal is running Linux.

3. Activate and use the environment:

```
conda activate phyluce-1.7.3
```

And well! Now you have Phyluce!

Practice 6. Installing FuSE

FUSE is an automated workflow designed to handle UCE data, although it can be applied to any type of sequence alignments. It integrates several steps to ensure that only reliable and informative sequences are retained. Here's how to download, set up, and run it in your **phyluce-1.7.3** Conda environment.

1. Go to your personal folder

```
cd ~
```

This ensures all files are stored in your home directory.

2. Download the FUSE code from GitHub

```
wget -O FUSE.zip https://github.com/rmonjaraz/FUSE/archive/refs/heads/main.zip
```

If wget is not available, use:

```
curl -L -o FUSE.zip https://github.com/rmonjaraz/FUSE/archive/refs/heads/main.zip
```

3. Unzip the downloaded file

```
unzip FUSE.zip
```

After unzipping, the structure should look like this:

```
~/FUSE-main/FUSE/FUSE.py
```

4. Verify the file location

```
ls ~/FUSE-main/FUSE/FUSE.py
```

You should see the full path printed. If not, check the folder name—it might still be FUSE-main.

5. Copy FUSE.py to the bin folder of your phyluce environment

```
cp ~/FUSE-main/FUSE/FUSE.py ~/miniconda3/envs/phyluce-1.7.3/bin/
```

6. Make FUSE.py executable

```
chmod 775 ~/miniconda3/envs/phyluce-1.7.3/bin/FUSE.py
```

7. Activate your Phyluce environment

```
conda activate phyluce-1.7.3
```

8. Test FUSE

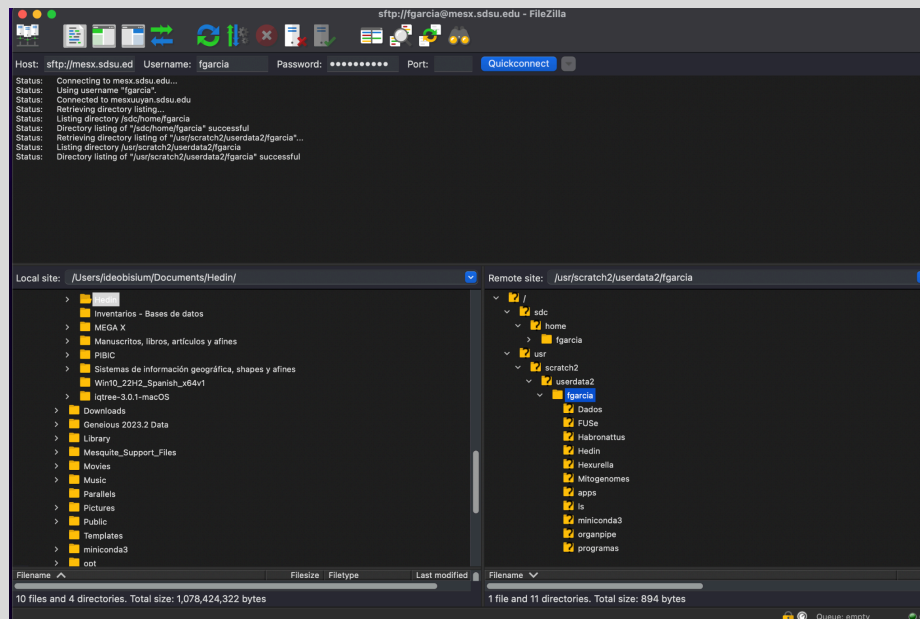
```
FUSE.py -h
```

This should display the help message and confirm that FUSE is correctly installed and executable from your Phyluce environment.

Practice 7. Using FileZilla

Sometimes (ALL THE TIME!) we need to transfer files from the server to our personal computers. This can be done directly from the console, but for us, beginners, it is easier to use a program that helps with file transfers. In this tutorial, we will use FileZilla.

FileZilla is a popular open-source FTP client for transferring files. It is versatile and efficient, making it especially useful for web developers, system administrators, and anyone who needs to move files between a local computer and a remote server securely.



How to Connect

1. Download and install FileZilla here: <https://filezilla-project.org/>
2. Open FileZilla.
3. At the top, enter the **Host**, **Username**, and **Password**.

For SDSU cluster, the host its: mesx.sdsu.edu and the port its 22

4. Click **Quickconnect** — and you are logged in.
 - **Local site:** your personal computer
 - **Remote site:** your server (Here you will put the address of your workspace on the cluster, check the Practice 2 : (/usr/scratch2/userdata2/fgarcia)

You will see a folder with your username on the server. To transfer files, simply **drag and drop** files between the local and remote sides.

Very well! Now you can transfer files from your computer to the cluster, or from the cluster to your computer. Use this in moderation, and for security reasons, do not transfer just any type of file. 😊

Congratulations! If you've made it this far, it means that all the programs are successfully installed and ready to use. You now have a fully functional bioinformatics environment with Phyluce, FUSE, Trimmomatic, SPAdes, and all necessary tools, and you are ready to confidently move on to the next tutorial,

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Take this opportunity to explore Phyluce, try different commands, and get comfortable navigating its workflow. Don't hesitate to create your own guides and tutorials for other functions or scripts, you can contribute to a broader, more accessible understanding of UCE data processing for the whole research community

References

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